

2022 Active Inference Livestream Review

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hello and welcome everyone it's December 31st 2022 we are here in the 2022 active inference live stream review welcome to the active inference Institute we're a participatory online Institute that is communicating learning and practicing applied active inference this is a recorded and an archived live stream so please provide us with feedback so we can improve our work all backgrounds and perspectives are welcome we'll be following good video etiquette for live streams head over to activeinference.org to learn more about getting involved in live streams or various other projects and learning groups at The Institute well today is going to be great and fun we are simply reviewing our streams from this past year we're going to touch lightly or mildly on all the streams that we carried out from 2022 and by way of doing that we may end up reflecting and preflecting thinking about where we're going to head in 2023 so we'll begin with just an introduction and a hello I'm Daniel I am a researcher in California and I'm excited to reflect this is like the dot zero uh preparation dot 1.2 discussion and then afterwards we reflect this is like the book and reflection for all of those and so it's like an even deeper level of nesting when we're going to juxtapose so many things that we talked about and I'm sure at some neural level whether somebody was familiar with the streams or whether they listen to none of them there's going to be something about just touching on so many topics and so many perspectives being juxtaposed that I think is going to be really Pleasant to you Dean yeah thanks Dan I'm Dean I'm here in Calgary um maybe I'm a revised or a recycled researcher I don't know what my title is now but um after Daniel said hey I think we should do this again I was like yeah we absolutely should because it's my metaphor is that it's kind of like double double frying those french fries it's uh it just adds an extra layer of integration or crispiness that's uh hard to resist you you get a because the way that we present these things is is hopefully a way of integrating a bunch of information because we Talk Amongst ourselves and it's helpful and then you look back at yourself talking about yourself and others and you just get a whole different perspective on it so I actually am really looking forward to sort of running through these things it's going to be a Sprint but that's okay it still gives you a chance to figure out what the Milestones were it's like seeing the entire Tour

de France in in 30 SEC or 30 minutes which is kind of fun it's like we completed that marathon and now we're high speed watching the recording from our um body camera all right well we'll keep with some patterns from the more paper-centric discussions like leading with big questions so what was our stream of attention in 2022 and where and how will we stream in 2023 what did you have to add on these questions um not not a heck of a lot I mean it's it'll be interesting given the amount of time that you committed to the Endeavor of here's 2022 I wonder what we'll fill it with I'm curious now that now that you're on the back side of that um what parts of what you hoped would be achieved came about and what parts do you think are still there's still room for some some back filling we'll come to that after we've touched on everything good yeah yeah I really liked what you added here though in a world that is becoming more familiar with congregation and digital mediums and spaces is our use of Technology more than an easy way to gain information access you could be in the library all you want and you won't read a book we can have all the access to information in the world and we won't necessarily digest and incorporate we can have all the generative AI in the world writing in these amazing essays and making art but where does that put us as information of forager and then you added can we turn figuring out into out there figuring and as more and more people start to want to figure out active inference it's not a Rubik's Cube on the table that gets finished and by putting it um out there figuring is where active infra ants are so I think that's really exciting like we do have so much to figure out from all backgrounds and yet we want to be out there figuring and having impact I think that's where your background in field research and my background in sending people out into professional settings yeah they're very different but also structurally very similar there's there's no substitute for actually picking yourself up and it's a form of migration I don't you like you use the word Stingray until I can actually figure out what my pheromone Trail is I'll I'll let you have that piece of it but I mean I understand it I get it and uh yeah let's see where it takes us all right roadmap so maybe we should have done like a a uh a rear view mirror or something but we had 87 streams in 2022 the live streams table has 87 rows so it was an amazing year and the numbers are shown here we're going to touch pretty lightly on all of the streams from two through nine the round table Symposium math stream or extreme Twitter spaces guest stream model stream and bookstream and then we're going to slow down a little bit and talk about some of the papers that we discussed so first the overall review though for a general Institute level review check out our quarterly Round Table number four so that's where we talk more about the institute's directions other projects and learning groups but today we're just going to talk about the live

streams all right what did you write with these questions yeah I'm I'm often curious because when you're when you're busy constructing 87 rows when when you reached whatever that limit is um is there a big picture that has come about as a result of that stack um again it'll probably be something we'll be able to answer better at the conclusion of our of our chat today but I think it's something to try to keep in mind I don't know that we can say with certainty that your big picture and my sense of that same picture is the same interpretation but it's it's a good it's a good idea to keep a sort of a general level sense of so did this all come together or you know how did the live streams intermingle with some of the guest streams intermingle with some of the bookstream stuff that you and um is it Tyler and blue are looking at right now well the links here can all be accessed through activeinference.org and of course one project of no is the active inference Journal where all of the live streams through the incredible work of Dave Douglas and others get transcribed archived and versions up on the GitHub so that is one structural development of the live streams is there a big picture it really can feel like just moving Pebbles back and forth and there isn't necessarily uh a vision of what the final nest looks like so I hope we can kind of look back on it once we've gone through it all right let's jump in yeah first round tables so we have a tradition of four quarterly round tables at The Institute scale the first two one will notice are active lab quarterly roundtables and the second or the third and the fourth are active inference Institute roundtables because in the middle of 2022 we had of course our transition from being active inference lab to active inference Institute so here's an image meme that we have from the fourth quarterly round table and these roundtables the screenshots with who's there and who shows up but also just what we discussed are like planting a series of flags in the grounds and they're an absolutely honest representation of who we were and what we were doing then and so I really enjoy sometimes even looking back and seeing how things have happened so that's not as uh scientifically or philosophically conceptual of a live stream series and the information in certain ways is out of date because it's more of an informative participation oriented communication but nonetheless a great series and also before we head in and to avert saying this of every single Series in the Stream big appreciation for those who had the bravery and capacity to join whether they joined repeatedly or one time it's been awesome and it's totally co-created with the people who are there in the most just unbelievable ways all right to the symposium so this was our second applied active inference symposium the focus was on Robotics and the first applied active infrared Symposium in 21 was just with Carl friston it was a one-man show and for this Symposium we had some more elaborate developments we had some external co-organizers in

Matt Brown and in Mark Miller we also wanted to host a wide variety of speakers and perspectives so we had two four-hour sections with one two three four five presenters in the first session and one two three four and a half with Briston joining in the second session and we saw the state of the art in active inference being applied to robotics we saw embodied actual robots we learned about some of the theory we saw connections to previous lineages of Robotics and also more General control theoretic areas each of the sessions closed with a Roundtable discussion the second one being more populous than the first and when people say like okay enough about what active inference is how and where can it be applied and people commonly point to areas like human health as well as robotics as some of the leading areas in the recent and in the coming years and I'm just really happy that we were able to highlight the state-of-the-art like this is what Robotics and active inference looked like in 2022 publish a transcript make it citable and get visibility across these different sectors so that they could connect on some of their developments and also their ongoing challenges math stream yeah go ahead good I hope that that uh at some point and I don't know timing really matters but I hope at some point people in other more dedicated learning spaces like formal education spaces systems will find themselves asking questions around how things that we've been talking about and we've been interacting on might complement the way that things happen in those systems so it's not let's go build a robot that does X it's actually taking the active and the inferred and taking actions on hidden States um as a complement to you know we've got to get a certain score on an SAT and I think it would be very interesting to see when that that um coming together on the horizon we know that the train tracks remain parallel but as we look at that there's a name for that you probably know what it is but when it when it reaches that Horizon it appears they've come together and so I'm hoping that um that isn't sort of Lost in some of these uh other initiatives that I know would actually be easier to do because there isn't such a history and and a whole a whole bunch of people who are very invested in maintaining the status quo as opposed to really journeying in ways that are um not discarding what we know taking the priors and allowing for some new priorities so I just want to kind of toss that in there awesome thank you we had just one math stream formally with David Spivak and blue facilitating on category Theory this is a few train stops away for some of us individually and for the field but this year there have been some incredible advances in active inference and category Theory so the field is being fertilized and prepared for an incredible regenerative category theoretic implementation of active inference but that's all we'll say for now here in the OR extremes we focus on the intersection of active inference and organizations we had

two streams first one was with Eugene Leventhal from the smart contract Research Forum and the second stream was with bishan hesri whose book governing continuous transformation is also the focus of the bookstream which we'll get to in a second and both of these were awesome streams there are hidden States for both all only say what I'll say which is that they both were incredibly informative and the triangulation of active inference for both of them was something that was incredibly natural um coming from as in bijan's case somebody with Decades of executive experience who completed his PhD dissertation on free energy governance and sees the framework specifically as a vehicle for moving the frontier of leadership in organizations or Eugene who's um fluency with internal and external organization made it a natural and enjoyable conversation and we connected a lot of topics of interest in online and web 3 so it was a great time yes that if you uh you want to do something interesting watch the work screen with um and then watch the live stream series number 36 about modeling ourselves and you'll very quickly see that um Bijan was closer to the end of the year but we were many of the things that he spoke of we were talking about in detail with one of the papers that we took up basically a year ago and that's the interesting that you can't you don't realize that until you're on a day like today awesome we had one Twitter spaces the title was can web 3 survive without cognitive modeling it was a provocative click bait headline and we had a chance on this discussion to talk about the active block fronts project with Jacob here as one of the primary developers and Advocates and stewards of that project and we discussed this question which is can web 3 or cyber physical systems more broadly exist in an increasingly complex present and future without something like cognitive modeling for which we believe active inference is the most composable and flexible generalizable and Powerful framework so I forget whether it's never ask a question in the headline if the answer is yes or no I I always forget so I never know so I always approach those headline based questions with an open mind but we had a good time in this discussion and these kinds of informal formats might have a lot of applicability we didn't really need to prepare much there also weren't any slides so for those who are interested in this genre of social audio or whatever it is called there's a lot of opportunities for frankly just interacting with people where they are talking about active inference and making the compromise in the improvised moment we had quite a handful of guest streams from number 14 to number 31. so I'm going to just briefly try to recall and mention those who joint Jorge estvest discussed osteopathy and mental health in a 14. Bobby azarian in 15.1 and 0.2 discussed Universal bayesianism Mark somes joined for 16 discussing Consciousness as Precision optimization which also was published by the active

inference journal in one of the most manually crafted and beautiful Publications spoke in guest room 17 about epistemic communities under active inference which is a area of increasing importance Ian Robertson Michael Kirchoff and Julian Kiverstein discussed the literal literalist fallacy and the free energy principle where that was only in March it was on Pi Day and we uh reawakened the Kraken of the realism instrumentalism dialectic which I know that we're going to touch on today again 19 Markov action discussed extended predictive Minds in guest stream 20 Shannon Proach discussed coordination dynamics of multi-agent interaction and the screenshot on the bottom left is showing a slide from her presentation I think it was an amazing presentation it's very accessible and very Cutting Edge work on multi-agent active inference and also in the very fun setting of music listening and creation so it was very engaging Adam Saffron in guest room 21 discuss free will Anna Lemke in guest room 22 discussed dopamine Nation and that's a fun one to view and share because Anna comes from a Psychiatry background not familiar with active inference too much but open-hearted open-minded and engaged with real people and with research every day so we had a really affable conversation in guest room 23. Maxwell Ramstead and Dalton's Octavio DeVille joined for rebooting the free energy principle literature which was a paper discussed that was not the title of a paper but a paper was discussed that got folded into the pre-play for livestream 49 so we'll talk more about Bayesian physics then Stephen Grossberg joined for guest room 24 explainable and reliable AI and also a commendation there to Ali who did incredible preparation and facilitation as well as editing the transcript here and made this an incredibly high quality experience for everybody guest room 25 Brett Anderson discussed the Continuum of traits between Autism and schizotyping so that was kind of interesting work and uh speaks to a lot of areas of meaning in guest room 26 along with my colleagues at Common Sense makers Daphna Shahaf Lauren Hebert Ronan Tamari and William Fisher we discussed from users to sense makers and about stigmergy which is collective behavioral algorithms that modify the environment in the context of sense making online guest stream 27 was with John Verviki and we had a great time we had a stool with three legs predictive processing relevance realization active inference and Dean and I were preparing those slides we had the stool going and then we flipped the stool and it was upside down and then that was just one of many inversions and Transformations and John has a very delightful and mindful way of approaching dialogue as those who are familiar with his work no and so we had a great time in 28 Giovanni Rolla discussed reconceiving rationality and inactive cognition guest room 29 Shauna Dobson on making up our minds imaginative deconstruction in math art for those who

are looking for the Triple Play of math art and philosophy there is only one unicorn in this Forest guest room 30 with keratinetrates was on the human governance problem complex systems and the limits of human cognition this was a very fascinating presentation of an artificial intelligence system that was churning in the background while we had a discussion and so it was quite visually and conceptually interesting and then most recently in guest room 31 Brett Kagan and Adil razi discussed biological and synthetic intelligence from the free energy principle perspective and some of their empirical work with neurons in a dish playing video games and the way that they were modeling that with active inference so a great set of guest streams and we look forward to a diverse and comprehensive set for 2023 so if you want to join for a guest stream or if you want to recommend or introduce somebody this is very low bar it can be a presentation they've given somewhere else some of these presenters requested us to do background which we're happy to do prepare questions others just show up so if you think there's somebody who could fit in well in a guest stream even if they're not like in the center of the active inference research development's ongoing we can absolutely make it happen 15 what would you add on the guest stream angle well if I could if I could invite people out myself at the same table and I could get Dalton and I could get Shauna yourself and John vervacci and me just sitting around and just talking about stuff I think I probably would be about as good as it gets I think that would be just one of the most amazing dialogues and improv you wouldn't know where it was going to go and that would be half the excitement is just people who who can who can basically have a conversation with anybody about topics that are incredibly sophisticated and yet seem to be able to bring it to a level where as you said it's engaging it's uh it's really it's amazing and this is not to downplay all the other guests that were kind enough to to join but again if if I were to if I were to uh do something on a social level on an informal level I all of these people thank you all to all of them great all right in the model streams we like to highlight modeling and we had three absolutely Cutting Edge presentations with interactive components model Stream 5 with Pietro masaglia and Tim ferbellen we discussed contrast of active inference which uses some current threads in non-active inference machine learning like contrast of learning and brought it into the context of active inference to facilitate learning in noisy environments in model stream 6 teal field Champion joined to discuss is quite prolific and creative Avenue of research into branching time active inference and Jacob and Ali are here also did a lot of background to prepare and as we entertain larger and larger State spaces and deeper and deeper possible Futures with more and more and more affordances and affordances on preferences and affordances on

affordances on preferences these State spaces become vast and things that you can write down on the back of an envelope might take all the heat in the world with current computers to run just like you could write down two large numbers and multiply them but you could look at each number and it's just a number well you can draw a base graph and yet to run that base graph might be not possible so there's a lot of relevance in this area of research of finding out these computational acceleration methods and heuristics as well as even just qualitatively this idea of branching time branching and pruning time and what kinds of strategies are used to branch and prune time adaptively it's very exciting and then in model stream seven Connor Hines and Daphne demekis joined to discuss Pi mdp which is rapidly becoming the python active inference implementation approach of choice in the community and it was great because we were able to ask basic questions we were able to hear about the intentions of the package and we're going to have a DOT two in just a couple of weeks this is something uh in terms of the package it's something where more and more of our work in the Institute is going to use this package and the interoperability of different generative models and processes and simulations written in pi mdp is going to facilitate the accessibility rigor and applicability of active inference so great work to all those who are contributors on this package and everybody who is out there applying active inference computationally and shared it with us this way all right and the last non-paper live stream is the book stream so the bookstream is being led by Blue and Tyler Solberg and when they got a hand on Bijan kesri's book governing continuous transformation reframing the strategy governance conversation they jumped on it and they said how can we do some kind of a regular series how can we increase our accountability in Reading understanding this book and also make something for the community and they said we want to do a weekly ish book stream so we've done that for the past several weeks and there will be many more sessions to come including discussions with the author so um great initiative it's keeping the stove warm on these topics of organizational development which are just absolutely Evergreen on to the paper live streams which we don't play favorites but we love the paper live streams go ahead Dean start us with 35. where were we in January 22 when we were reading a tale of two architectures free energy its models and modularity by Majid Benny I wish I could say that I looked a lot younger back then but I looked old back then too um I think a year ago we were pretty excited because I think it it felt as though things could go in a number of different directions and that we as a community that were discussing a lot of the you know most Cutting Edge on the sort of bow wave of information that was coming out of people's research it kind of felt like we weren't just aggregating we were actually kind of

one step removed we weren't getting too locked into any one particular facet of this multifaceted idea and uh it could it could um it could really blossom into something like we had we had this really what we thought was this kind of magic Acorn or unicorn or whatever and how it would be viewed um 360 some days later we didn't really know but we also weren't afraid to kind of test the limits that's what I felt 22 was more than anything else it was an opportunity to test things and that didn't mean that we were out to prove people wrong we just wanted there was now enough information and not a large enough sample size that we could do some cross-checking and that sort of allowed the people that were doing a cross-checking the ones that were reading this long list of papers a chance to kind of go huh okay well I think I've seen that over here but there seemed to be some hesitation around whether that how far that rule holds under this set of circumstances and yada yada so 35.0 started out I've just put my notes here as a formal treatment of what graphical modeling can look like I.E are dag dags are cyclists well DCM are cycle based and for me the introduction of the term joint probability up to that point I'd never heard of that expression and we ended up using joint probability in a lot of the papers that followed so there was a bit of you could start weaving Golden Threads between things Majid who eventually joined us for this um nothing but props for that guy for again pulling in what are seemingly really distal Concepts and making us realize oh well if you do do that you gain Insight pretty quickly then we went on to 35.1 which was which do we start with representation and or effect both in order to model uh so recognizing the importance of personal information encapsulation that everything doesn't have to be absolute especially over different time frames not just architectures although that term was in the title of his paper uh but the role of the architect someone who is an applied structurer if there is such a thing and the sort of arbitrary nature of that that's how how does it remain arbitrary well by not being absolute a real clear way of understanding the difference between function and effective connectivity that took up a lot of our conversation uh proximal consistency through time and introduce the perspectives what the term perspective swaps I didn't realize that I'd introduced that into the conversation um but yes that's where it all was seeded was back in January almost a year ago then 35.2 what are we trying to trace separation and or boundedness the answer to that as Majid clearly concluded depends on what's available so availability not just our regime of attention so I thought that was an incredible thing to bring up a year ago do we keep that what are what that we are the trans the tracers and the modelers and the separators and the bounders in the front of our minds as we analyze situations if learning is decided sought after effect when does that help does this lead to a year of holding up when we

are looking at a Target system say active inference is subject and when we as observers might be using active inference in formal ways I.E or allowing priors to not get in the way of new insights assuming insights are a priority for agents and I was interested in thinking I wonder how they would be able to get that into JS robots right so that's an interesting thing to me this is the Divide between being a programmer of a robot where the priority is learn uh versus dictating how priors are amassed and organized and then inhabiting a programming or participatory experience one where deconstruction is adaptability until ultimately something called complexity dominates the conversation in both of those situations priors and priorities work in concert but in the latter the humanist cases are on opposite sides of a Markov blanket so at least partially hidden but also partially encapsulated we talked a lot about duck rabbits that are not riding to unicycles concurrently whereas the robot might that's an interesting difference there and for humans riding two unicycles is equal and complete partitioning or encapsulation so that's why it's hard to do takes much training to be able to carry that out so that's 35s that was Majid and uh bringing the the science and the modeling and the abductive and that was our conversation even though his paper was in response to a couple of other authors um arguing that there's only one way really to model and so again lots of things grew out of a paper that was meant as a response and created a whole different ecosystem awesome wow I'll just give a short thought graphical models and network Theory are at the heart of active inference as well as many other fields and what it means to have a network or a graphical model is to represent a system or a model in terms of nodes and edges and exactly what those nodes and edges are matters a lot and in this diagram we had the parable of Bob and Alice who brushed their teeth at a similar time because people in that region just happen to do so and we discussed how the statistical connections between their activities in terms of the functional and effective connectivity are not the same as what is touching what in the real world and this very straightforward deflationary approach to separating the map from the territory yep came back again and again and again it was like we were on the Carousel of and the map territory and the math is not the territory all of those topics are so Central because when we're looking at those generative models we're looking at a graphical map not a picture of the territory and it's easy even when you know to forget and again I I often wonder how because Daniel reminds me no people vote which papers we're going to do and sometimes it's just availability of authors but the next paper which was 36 could not have been a better Co uh what's the word co-inhabitant the with with majeed's work it was uh like the perfect follow-up Matt and Giovanni's uh work on on modeling ourselves yep give an overview go for it uh 36

was the importance of working backwards you're the way that you and blue set up the the entire paper was well we're going to go to the end and we're going to come we're going to work our way back through this paper just so that it makes sense because I think the there was a bit of a woo-woo kind of uh sense from reading the paper because it was very very deeply philosophical um also just introduce the claim that in order for internal representations to be wiped out we need to marginalize the unit call in action really radical as if adding an exclamation point to your claim somehow changes the relationship of an action which is a part to the entirety which is a whole so you when you guys first picked up the paper and you're doing the dot zero um already there was like okay well I wonder what we're getting into here well you also asked indirectly how artists indirectly how artists people with good external rep who are good external external representationalists maybe have a different gapping or way of entangling or both the organizational structural and content related and functional aspects is either representative or non-representative within parameters uh yeah a nice table there set up which boy you got a lot of shelf life out of it who is useful for all three live streams what I found most interesting was that to be an activist meant to you the use of simple heuristics means control mechanisms do not need to get to the level of sophistication more than what uh what short catch and long you had a great diagram of the baseball Carl hitting it out of the park categorizing mine the fep equals you can be all over the map does that mean we can engineer a bridge to everywhere which was one of the questions I'd written down actually after watching you you and blue do the dot zero then in the dot one how does transition go from materialize to non-materialize and then back and that was that brought us to the idea of what stable and what's dynamic juggled as a process of predictivity in order for representation and non-representation to coexist this was the great thing they are they are bedfellows Dean and blue debated the scale freeness and the scale friendliness of science I.E observations recordings analysis and prediction so modeling ourselves in real time with Daniel's uh substituting disciplinary for generalization as modeler strategy the question was raised around what must exist pre-plan pre-model and then we I don't know the lady's name who joined us for the Danielle Danielle we got into some really interesting loops with your um table again we started we were looking at organizational structural and content related has probably more stable things that we want to Anchor with those edges and nodes represented and not kind of moving around all over the place or without edges you can have a gap like but there there was a point that came up was the black lines separating the rows and the and the um columns were they solid or were they actual gaps so that was kind of interesting um uh and then there was a

functional as radical Dynamic and dependent on those stabilities uh now is modeling ourselves how when where what what if and which if we act how does that impact the model that we believe exists as we boundary cross which again if you take it to the stuff that vision is talking about on a on an organizational level that gets real that's combinatorially explosive so it's interesting that the how we model has a huge impact on how do we model ourselves as an organization as a weave that which we'd like to remain constant with that which we'd like to see changed so that was uh modeling ourselves and again it was Matt and Giovanni didn't show up but man would have been awesome to have them there and kind of lead that circus because it was great it was great debate between blue and I because she really wants science to be seen as it and I really like science to be seen as specific and you are the great moderator and said no it's both so I was 36 bottling ourselves wow yeah I totally forgot about Carl hitting it out of the park about Carl bonds that is fantastic hitting the particle out of the park and and about where is representation oh representation where art thy staying yeah that was it was too fun it was too fun and and also just so people um know like the live stream page has the slides for every um every single live stream that we're discussing you can find um you can find the live stream the title the paper the transcript the GitHub where you can contribute to the journal as well as the slides so when we're pulling slides you can also look at them so yeah this was 36 was awesome all right 37 37 free energy a user's guide by Stephen Francis Mann Ross Payne and Michael Kirchoff from 2021 you want to take a first pass well I just I just remember that uh as much as I'd like to be able to say I could do the math um if somebody else does the math and I can follow along at my own pace this is where I learned Jacob's gift basically so that's what my first pass is I know what I know when I'm when I'm I'm in the presence of people who could go a lot further on that plane than I can so yeah yeah the the paper is an Excellence user's guide or introduction one memorable section was on this trilemma or trielectric of general or philosophical claims mathematical claims and empirical claims and they went through Edge by Edge and they asked how do claims from each of those three vertices justify each other and so we hear that all the time people speaking about active inference and other areas well because the brain is this way we use this equation or we use this equation because the brain is this way or because of this philosophical belief we use this equation it's like seeing it laid out that way brings a huge amount of rigor and it also helps us understand that rigor is not just increasingly technical math rigor includes the relationships among mathematical philosophical and empirical claims from different fields so that was one key takeaway for me and another was this um the three piece the tale of three p's and it was a focus of the paper to

point to the unconventional way in which preferences are utilized in active inference which is not as a reward or a goal function but rather something that is kind of the fundamental frequency of the generative model and so you have entities striving to return to the bottom of the bowl in terms of the path of least action in terms of the frequency that they want to be at and that is the embodiment of preference and as we came to see later homeostasis and allostasis but there isn't some sort of dual track between what is and what should be rather there's a bowl that represents what could be and is and should be through its shape and it's about the ball moving on that shape by which preferences come to be enacted so that was something that was pointed to and very approachable and they talked a lot about how they situated the audience of the work in terms of wanting to speak to philosophers who really did want to learn about what free energy principle is about one of the things that I got kind of reinforced in my mind is because I I kind of believe that Friars are on one side of a Markov blanket and priorities are on another what that Tale of Three P's kind of in my head gave me a little bit of latitude on was preferences can be on either side of the blanket they can literally cross through they can they can pair out preferences can pair up with your priors you can figure out what those things are that have been sort of burnt into your your neuro Nero desire lines and they can also preferences can also be working with with your your priorities in this moment right as you find yourself now okay this is what the niches the information the niche is sharing with me but what's my priority what's my preference as well so that was what the the three piece thing I remember now when we were talking about that's going oh okay don't get the three things mixed up give each one of them their their time in the spotlight um all right live stream 38. was evolution of brain architectures for predictive coding and active inference by Giovanni pizzullo Thomas Parr Carl fursten from December 2021. and the topic here is so important what is the evolutionary neurophysiological basis of cognition and how to complex cognitive phenotypes arise at a developmental time scale at an ecological time scale in an evolutionary time scale and where this paper eventually takes us is into thinking about the graphical models again remember what we discussed just a few slides ago about graphical models the graphical models which enable or facilitate or underlie different kinds of functional outcomes and then how structural modification on graphical model architectures enable different traits to emerge diverge duplicate and innovate and so here we see on the right side it's a a tree of options starting from a relatively simple initial graphical model how different kinds of modifications duplications Etc can result in the kinds of functional as well as structural phenomena that we observe in living creatures today and whether that's the

redundancy of Gene regulatory networks or the paralyzed architecture of the mammalian cortex these are the kinds of phenomena that this paper set out to bridge the gap with between simply representing things as they are as graphical models and understanding what if you weren't just studying the ant brain today but what if you were studying the amp brain over its phylogenetic history how would those generative models be changing how do things like temporal depth or Intermodal Fusion come into play and so that was a nice paper that connects to Evolution and a bunch of other behavioral areas really helped me understand or confirmed in my mind that there's a big difference between a person who makes a model and somebody who has to interpret somebody's third party model because I remember going through this I was I felt like I was learning on the Fly because even as I was reading the paper there was a lot there to try to parse and just having to be able to converse with you in blue and I think Stephen was a part of this one as well for a bit of time it made it a lot easier for me to kind of understand As an interpreter of models as opposed to somebody who's here's my model what do you think of it it was taking some of these really complicated models that other people have designed and then trying to put make it into something that I had adopted nice okay live stream 39 morphogenesis as Bayesian inference a variational approach to pattern formation and control and complex biological systems by Franz kuchly and Carl friston Georgie georgiev and Michael Levin this paper took what in many other situations are a mathematics of structurally stable but dynamically signaling architectures like the brain and considered the case of structurally volatile dynamically signaling structures and so they focused on one of the key model systems in the Levin lab which is the planarian morphogenesis upon cutting this animal it quickly reforms into a self-similar body plan so that has implications that are being developed in a whole host of areas related to health and development and disease and the paper uses a somewhat technical variational Bayesian mathematics which we also jumped into and we had Franz join us here in the office overlooking some gray featureless landscape and had just incredible um insights to share and we talked about the environment what our developmental priors and trajectories what are preferences over morphologies and this really opened up the discussion from as mentioned at the beginning systems that have a structural stasis that do signaling in that structural stasis like all the computers are hooked up and now they're talking to each other but they're hooked up the way they're hooked up well what if the computers can change how they're hooked up or what if they're moving around that is another space that also connects us to a lot of the physics of the embodied worlds maybe there's friction maybe there's viscosity maybe there's a razor that splits two halves from each other

and um it was a great discussion to see how the mathematics of the variational Bayesian approach is able to accommodate these different focal systems which speaks to the generalizability of active inference there are a lot of formalisms there but it didn't feel normal talking with Frank and it just felt like the possible with regeneration we had had a whole bunch of fields that yet to be explored and the kind of openness thinking openness that that opens up possibility was really that was really cool about that live stream set awesome all right 40. who could forget a free energy principle for generic Quantum systems 2021 paper in fact a year and a day ago by Chris Fields Carl Friston James Glazebrook and Michael Levin so this paper we needed to situate ourself on one hand here represented as the x-axis sometimes we find ourselves going back and forth between more talking about free energy principle and active inference other times discussing literature or ideas that are outside of the free energy principle and then orthogonally we can think about this Continuum between not Quantum and Quantum perspectives from physics and we were in the upper right quadrant or another way of saying that is we were everywhere because we were Quantum and non-quantum and we were in fep we had the authors join for both of the streams the 40.1 and 40.2 so here we are with blue and Steven us and and the authors every sequence of symbols that these fellows added was immense and to be able to make that Daredevil jump across the Grand Canyon and connect the most advanced perspectives on quantum mechanics quantum physics Quantum computation unconventional computation more broadly for example thinking about the perception action and memory phenomena in terms of repeated measurements of a Quantum Observer with their reference frame and looking at the holograph it was a great and memorable sequence it it literally materialized the bridge to everywhere being in that upper right quadrant which is which is a fascinating thing because I don't know how you engineer Abridged everywhere but you the possibility means you have to open up to more than somewhere and I think that that was again one of those one of those sets where you're kind of going am I really doing well you wouldn't say this but I would say this am I really doing this am I really having a conversation am I get do I get to sit at the same room with people of this capacity it was really fantastic and I got to talk to Mike Levin for privately after that because uh yeah the idea of prediction matter expertise intrigued him so that was very cool great wow all right carrying through 41 extended active inference constructing predictive cognition Beyond skulls by Axel Constant Andy Clark Michael Kirchhoff and Carl Friston so here we had [Music] we had the All-Star team we had blue you and I trust is Like Glue if you don't have it you won't be able to stick together and our team talked about trust and active inference and I

remember this section of the discussion where we had two systems the blue and the orange and we discussed like what is their border what is their interface is it another region but then if it's another region isn't there a region that borders and what is on the outside of that and so how does all that connect to Extended cognitive systems extended relative to what and so this topic of extended inference was dealt with thoroughly here and the mathematics and the applications of extended cognition are so promising I also remember you discussing the difference between a type of active versus a passive offloading there's a difference between somebody writing a code on a Post-It note that only they know so if they look at it later it's like it helps them remember what they were thinking of but to somebody else it doesn't mean anything and it doesn't do anything as just notes on paper versus JF creating a robot and sending it into the environment yes it might still need recharging and so on so it's not like it's totally disconnected but it's doing things actively and that kind of active passive offloading and the way that that facilitates our cognitive goals was very rich discussion and I I actually being the first author of this he always to me he's like Majid he he brings in the influence and the how long do you stick with something before you you no longer you know can carry something on what is that what he's he's so he's such a good explainer of you realize that the environment is actively inferring you as much as you are inferring it that's that's really hard for people to get their minds wrapped I mean the first time you hear that you're kind of like but he does such an amazing job of explaining that it doesn't it doesn't seem like people talking about some silly thing like fairies dancing on the head of a pin yep you actually I go oh wait a second you're right it would have to I would have to the environment would have to be scrutinizing me as much as I'm scrutinizing and trying to figure it out nice all right 42. robot navigation as High rock hierarchical act of inference by ozankatal Tim verballon tune Vonda melee Bart Dot and Adam saffron this paper was an incredible example of Applied active inference they in this paper represented the full rainbow from conceptualizing a generative model to implementing it and making a robot that actually enacts that generative model in a warehouse setting and what that warehouse setting required from the generative model was that there was two levels at which the generative model was playing out at the higher level which we were thinking of in terms of strategy the robot decides where it wants to be where it prefers to be and where it is at a lower level which we connected to tactics is related to the kinds of bodily moves that it can engage in it's more movement affordances and this discussion facilitated a lot of action into the math the computation the embodiment the Robotics and give some of the clearest examples of Applied hierarchical modeling that came

back again and again and again so that was fun 43. paper 43 was predictive coding a theoretical and experimental review by Baron millage and you'll Seth and Christopher Buckley this paper was quite a doozy it was very mathematical and quite long the memorable part of this paper to me was collaborating with Maria on the dot zero Maria's philosophical knowledge with predictive processing took us way back and that was rooting the discussion in the ancients thinking about World Knowledge traditions and Kant and helmholtz we hear a lot more about that kind of 17 18 1900 philosophy time but this pulled us temporally back and helped Bridge the math of this paper which was technical to the ways that people have simply perceived cognized acted and thought about the world around them and then to speak to the math this paper almost the entire thing before something like the last several pages it's about hierarchical predictive processing architectures but there's no action to be found it's just about inference inference going up and down priors and observations and prediction errors all of that and then at the end oh yeah we're going to be doing inference on action we're going to be doing active inference and with that tiny Twist of the knife active inference becomes and so it's just amazing to see how purely inferential architectures when they're doing inference on action the consequences of action are taken into account policy selection is treated as a formal thing to do inference on you get active inference and it really isn't too much more complex than that and then just a last point just like we had the three areas of claims justifying each other from live stream 37 with the general claims mathematical and empirical claims we had another triangle in 43. so here we had on the top biological systems and what they were doing on the left we had philosophy and qualitative claims and on the right we had formal quantitative claims and so we made another triangle with pretty much the same three vertices which is mathematics philosophy and empirical biological phenomena now we took another first principles pass on constructing those three areas and what are the nodes what are the edges and how our inference and action at the heart of all of it 44. 44 was the Therapeutic Alliance as active inference the role of Therapeutic Touch and synchrony from 2022 by so McFarland Francesco cerritelli Carl friston and George ESPYs so this paper was awesome also here from the screenshot of the event we have Ian tenant with the inner sense Channel and projects who saw this paper brought it to our attention said I want to facilitate this discussion and I want to connect interception and the Therapeutic Alliance in a modern context to active inference and I want to bring in these awesome authors and make this a conversation that really hits home and so thanks Ian for for joining in and helping that way the paper focuses on the Therapeutic Alliance which is like the clinical or medical or therapeutic niche it's the

people in the room it's the receptionist it's the nurses it's the surgeon it's the physical rehab practitioner it's the family of the patient it's the patient themselves it's the physical surroundings and to be able to very broadly discuss that vital setting while connecting it to Key active inference topics like generative model synchrony and so on as well as to see their delightful figures and the way that they represented the same a bunch of authors joining for both of the discussions it was a awesome series this one 45. this is what I'd come back from the dead and this is what I walked into I knew there is a reason why I wasn't called away it was because of 45. 45 was the the free energy principle made simpler but not too simple by Carl furston Lancelot DeCosta nor Sajid Connor Hines Kyle olholtzerver grigorous padiotis and Thomas Parr well I think the best way to summarize or crystallize the experience of this stream was blue and I spent many hours preparing for the dot zero we were having separate meetings and doing asynchronous work to really work through and understand the mathematics of this paper and we did the 45.0 in May it was all great we were looking forward to June um we had invited the authors we didn't really know what was going to happen and it was several minutes after the hour where we were supposed to start and so we usually just try to start on the hour or a few minutes after the hour why wait longer and we just didn't have anyone else on stream so we said all right this is it which we we floundered through the dot zero let us swim again into the dot one yes please Dean I'm sorry but you're not telling the whole story when when you are completely out of your niche and you guys worked so hard to make sure that you could adapt for this but then you took that you had the courage and the strength to literally jump out of the ocean into the boat and then all you wanted was come on it would be wonderful if somebody could show up and at least tell us that jumping out of the ocean was a good thing to do and then just like that yeah yeah just a few minutes after we started come along and say hey how's it going here catch and release it was so amazing and I mean you you might may not believe that it was that big of a deal but it was and it was amazing yeah just a few minutes after starting with the two of us first Carl first enjoined then several minutes later Thomas joins yes it was one moment but it spoke to putting it out there yeah and being ready to show up and being ready for others to show up but also for these two sessions that both Carl and Thomas joined for those who want to know about the mathematics of active inference yeah there's there's many miles to go with making it accessible rigorous and applicable so not to be denied but if somebody wants to know the absolute state of the art these conversations you're hearing it directly and the questions and the discourse are 100 approachable so this was just some of the highest density learning and Higher Learning it it

was an Institute of Higher Learning at that time if you want it if you want evidence of a naivete which is not something you want to remain as all your life but those moments when you allow yourself to be naive and how that can be a feature watch this if you want to if you want a first step you might look at going I don't know anything about active inference just still you may not understand some of the things that were discussed and that doesn't matter what you really will appreciate is the generosity great all right 46. so 46 was active inference models do not contradict folk Psychology by Ryan Smith Maxwell ramstead and Alex Kiefer what is there to say we are folk psychology we are them and this is also a topic that people bring up all the time oh preference are we talking about like that I prefer to have this happen in my life belief are we talking about like what it feels like to believe something and this paper just absolutely squarely targets those questions the conclusion is in the title yeah for the authors it's not a reification of folk psychology it's not chapter two on folk psychology it's not a nullification it's not a contradiction it is a non-contradiction right which is exactly what it sounds like which may infuriate those on any side of multiple fences it was a great conversation because the topic was so directly tangible and accessible we did want to connect the math and formalism of active inference to how people talked about folk psychology which is just like us day to day talking about our feelings so that was an amazing context and also this sort of fun stream memory was I used social media to uh communicate that the dot zero was beginning I guess on June 14th here and Ryan Smith just says I don't know if I'm going to be free later but I'm free now so we had Ryan in the saddle for the Dodge zero which was the the first time the first time we've ever had an author join and bring that kind of insight into the dot zero so it um you know class was held so yeah we were Court was in session and Ryan has joined multiple times and and has so much incredible Clarity on the math and on the psychology and the applications of active inference anyone studying his publication or presentation career can know that so it was a great conversation with Ryan and then also um with Alex Kiefer joining in the subsequent discussions and um Dean you brought so many fun images and motifs to bear it was just like it was such a fun house well here's the thing and again this was by June right we started in January trying to figure out what did the lines between mean right so is is active inference a divider is active inference an enveloper what this paper said was it's not contradictory to appreciate that it could be both that we have a circle you can be inside with the spokes or you could be outside and be the the contact service if it's a divider it can be on either side of the line but if it is both why would you always feel that you're doing something wrong by holding up both right why is both the contradictory

aspect of this which again encouraged me in my men too I don't care how you want to do it you can collapse past that but if you want to get to fractalization fractalization being the moment that you're literally in the gap or on the line or on the boundary right that's what affords fractalization is the fact that you don't necessarily go well that's a contradiction Hands Clean we're done no I I have to stick to something simpler because I just I will not accept a contradictory space there are times for absolutisms but there's also what Ryan was pointing out which is that maybe that's a feature maybe this this ability to see both is something we should just not you don't have to accept it but what again counter factual what if you did so it was really good and then Alex showing up afterwards kind of playing cleanup it was really cool yep so that was taking us to the end of June we then took a few weeks off of discussing papers and in that inter-semester break yeah we came to 47 so I'll let you describe 47. it was octave inference and abduction 2021 by ativeko pieteren and Majid Benny and an active Dynamic social cognition and active inference by anes hippolito and Thomas Van Ness 2021. yeah well again again there was I guess there was an intention on my part to get first of all muster up the courage because these people have spent a lot of time putting these papers together um but my sense of it was I've got both eyes open I'm not focused on instrumentalism or an action or taking a realist stance or taking a non-realist stance and yet there was all of this conversation around those things as to which one entails the other or which one is came first and all I wanted to do was hold the papers up side by side and rather than having people having to defend one side or the other I wanted to hold it up as here's two sides to the same coin we're on the edge of this coin now looking down on on both and what does that imply that's essentially what that was about because I think at the core or the heart of prediction matter expertise not just active inference as something that we can subjectify but that we can act upon in formal ways not just informally that's that's when you know you've entered the the you've entered the divide or you're on the boundary which leaves it up to you as the person now that's taken up those coordinates to do things like we're doing today reflect back on what got you here was it one leg or the other right it's like saying oh well it was my left leg that got me here well it was but it was also the right leg so how do you square that in in you know some of these debates where people's like I said they invest a lot of time and energy and effort into trying to make sure that what they're saying is correct to the point where they almost feel like they have to defend something because it narrowly fits and I wouldn't argue against that but if we're talking about both scale free and scale scale friendly which we are I think we've all kind of accepted that now um then how does that work in machine despite the fact that he was not under under

the most Optimal Health conditions was super kind showed up and once again graced us with trying to explain some things despite the fact that I know he wasn't in the in the best of condition to do that so again another um shout out for Majeed Benny and his willingness to go the extra mile yep and for you in Breaking the one paper per series speed limit and having us consider two papers wholeheartedly so that was completely it was helpful that's all I was hoping is that you know it wouldn't be seen as as being disrespectful because we weren't giving enough attention over to something we were actually entertaining availability with attention again we were men men doing things and um yep that was cool yep thank you thank you communication as socially extended active inference and ecological approach to communicative Behavior by Ramitison and Pierre Poirier this paper brought together a lot of the topics that we've mentioned earlier communication extended cognition social um and we had uh you blue and I with Remy both times what can you say Remy's Remy is a great defender of an a fantastic defender of an accident and he also has partial encapsulation whether he would have described it as that or not he is a partial encapsulator there you go yeah it was um it was awesome to have him on and the open-endedness of the conversation as we took a look on the social which can get one lost from the first moment and maybe that's not even a bad thing but we we lost more than we've ever found in that niche of exploring And getting into the last three papers here yeah and the reason why I mentioned the gap between 46 and 47 was because the we came back with a fury we were mad at anyone we were angry but we were um Furious and we uh we had some incredible streams so yeah almost Center in representing that for me is in live stream 49 a worked example of the Bayesian mechanics of classical objects by Dalton Saktiva DeVille so for this paper Ali Jacob and our colleague Ander we met weekly for about two months to prepare for this discussion because this paper and more generally Dalton at all's advances in merely 2022. have transformed the status of active inference in the free energy principle if you want to learn more about what that really means check out live stream 49. Ali Jacob and under put in a ton of work we worked as hard as we could leading up to the dot zero and then Dalton took us on a journey in the dot one and the dot two we couldn't have asked for a better elucidation or a more enthralling presentation of Bayesian physics I really mean it and whether you're working in classical thermodynamic or quantum mechanics or whether you're not familiar with those areas this direction of work and specifically the presentation and the answers that Dalton provides and just the spirit that he brought are our absolutely memorable so I'll just give you a little aside this is the only time on any live stream ever where instead of playing music when I'm doing my workout my third

lesson my third lesson of this the the dot one of 49.1 was while I was working out because it's that good I could act because it had to be my third listen because the first two listens I was just like having to pick my job off the floor but by the third time I wanted to listen to it a third time and instead of listening to music I was listening to you guys work through this stuff and I was just like this path integral stuff changes everything yeah just briefly for the for the conceptual here are the three faces of Bayesian mechanics using a system and a scheme that's presented in this research and an earlier research by Dalton and Maxwell at all one setting is where a ball is resting on the ground and it's not moving that is kind of like a distribution a belief distribution that's getting new input but it's not updating it's just getting the exact information it expects it's not updating it's not moving in contrast when new information is provided the prior updates to the posterior and it moves to track the mode of the new Optimum so that's like if you had a ball on a staircase then you pulled one stair and it fell down to the next one and you pull the next stair and it fell down to the next one those both represent taking densities Over States like preference as a State observation as a state states can change through time but they at any given time are State based in contrast and that's where we have the non-equilibrium steady State Ness density so all the discussions we had about non-equilibrium thermodynamics and non-equilibrium steady States all of that Ness density discussion is on the states what some of these recent developments in path integrals gauge Theory caliber Theory maximum Cal as well as the Dual the discovery of Dalton about The Duality between the constrained maximum entropy principle and the free energy when we calculate densities over paths there does not need to be a Nas assumption or a discussion of a mode but you can get path tracking a path of least action tracking without appealing to this intermediated Ness it's only been a few months but there's so much more to come here on the mathematics and on the applications it's just it's incredible and a year ago we couldn't have said any of this so a lot has happened in active inference but in the 49.0 you'll see us grappling with the eras of active inference and fep and about what phases in epochs and even now we can acknowledge that we're in a post-daltonian era yeah and and it's funny because I before hearing this I it almost felt like you had to go down the vertical and then across the horizontal and then down the vertical and then across the horizontal and then all of a sudden he introduced the wheels and the bike went riding down the steps that's how different it it became now that's still a bumpy ride you have to go quite quickly to smooth things out right smoothing right but that's that's essentially how dramatic it was like it was like introducing the wheel it's only so it's only a wheel right Wheels within Wheels yeah all right 50. so in 50 the paper being discussed I didn't bring it in was um

interception is modeling allostasis as control with Eli sanesh and Jordan Thoreau who are here as well as several other authors Karen quickly and Lisa Feldman Barrett maybe some others so what what would you add on 50. well just that again you would see that interception would be its own field and allostasis as being a separate field and then there's this lab in somewhere in Northeast in the United States that's trying to defield bring things into a an opportunity to be integrated and it was fascinating to try to see how people address operationally and formally things that seem to be pretty messy and hard to manage when we're talking about things like we don't have necessarily access to the information in a mathematical way it's a it's called a Feeling so how do we how do we blend those things together and again I'll be honest I was the most skeptical and became a fanboy after having great conversation with Jordan and Eli yep that was a memorable discussions Eli gave a presentation that helped us really like lock in conceptually how they were framing it and what they did some of the statistical Transformations are evergreen in the sense that there was a capacity curve a probability density implied by a capacity curve and then a responsiveness that was a log probability density and now whenever we see one of these we see all of them right because they're all just Transformations either PDF to CDF or log probability densities check out the stream to learn more but whenever we see these we can kind of see all of them when we see one see one thing see multiple things learn one thing learn multiple things yeah it was awesome all right and the final live stream discussion paper so this was uh canonical neural networks perform active inference 2022 paper by takuya isamura hadiaki shimizaki and Carl fursten and this was uh it was a it was a fun and an honor and a Learning Journey for me I prepared the dot zero myself which hadn't happened for a little bit of time by then um this was just in November and then uh in both the dot one and the dot two it was just takuya and I and the papers implications are so stunning in light of the prevalence of neural networks and artificial intelligence machine learning the paper supports a claim that was latent within an earlier research paper by the authors but they kind of completed the loop in this paper to really uh reinforce it that as per their figure one neural network canonical neural network listen to the live stream slash read the paper to learn more every neural network in engagement with the environment through perception cognition and action can be understood using the particular partitioning that we use in active inference where the generative process engages the generative model of an agent also in terms of perception cognition and action in fact the training by which agents can learn from their own experiences about what to do in the future is also analogous as well and here we see some message passing reflecting that the discussion was technical yet asked basic questions

everybody who works with neural networks or knows somebody who does I'd recommend that they check out this discussion and paper because it was a perfect example of like making the bridge and here we see like the the Min 2 of Min twos we see the partition between the generative model and the generative process and then two views on that being juxtaposed and so now when we see a neural network we can think that there's a generative model that might reflect the semantics of what the neural network does even if the structure of the neural network is Arcane and in contrast when we see a generative model we can imagine that there are some neural network architectures with Arcane topologies that Implement that generative model and so as we build these rigorous Bridges we see more and more with the same visual input and so that is really advancing the field and also again could not have been said a year ago isn't building a bridge to everywhere or is it getting is it moving in that direction let's go to Reflections oh okay if we built Bridges to everywhere it's like paved paradise put up a parking lot you know we wanted people to be able to drive all over paradise and park wherever they wanted so we paved it we put up a parking lot I don't think that's what's happening okay what is happening well that's I keep I mean again I I get the ambition of trying to take some statistical models and see where they might be inserted in things that are going out and doing the exploration see that um and my where am I here because I my continuing attention to active inference is because I believe it is a way when you're ready it is a way of generating way more insight way more aha moments than what I think is a kind of unidirectional approach or a hierarchical approach that says Mass all this information put it in synopsise it into your own synthesize it synopsise it and feed it back to a um a board or a Review Committee or or something like that I understand how that works and why that serves a certain agenda or or or intent I think what active inference test is it opens things up so that you do not have a ladder to climb or at least they haven't invented a ladder to everywhere I think that's what it does is it it creates options and possibilities and sees things as it could be this it could be that so from a humanist standpoint it just makes available it provides affordances that's kind of at the core of the work that's being done what it it it enables the use of things like counterfactuals why that matters why it's just as important to ask what if as it is to ask what is these are all things that I think don't even have to be stuck in narrative spaces I think it's it's what you now can choose to try and make some sense out of and I think a lot of times in order for people to become Specialists and experts at something they have to really focus their attention on something but I think what active inference says is the more times you vary your attention and change your attentional space I.E make yourself available to ideas and phenomena that

you otherwise would ignore because you think it's it doesn't really matter in this particular moment the more ahas the more insights you yourself gained so I don't know that we're Paving over anything I don't think that the bridge has to be something positivist and material in fact as we talked about at that very second set of live streams this year is the thing is the hidden State hidden because of a shroud something that's material something that blinds us to the situation or is it simply something that we just haven't given our attention to over yet it's it do I need do I need to um build a bridge across the Grand Canyon or is seeing across it to the other side enough that's what active inference asks us to explore um I think if people want to come and and find an opportunity to socialize with other people around references there's work to do that to do with that around stuff that Bijan has written about and stuff that Carl has written about and those are fantastic exercises but I think there's more to it than that and I think the everywhere part is something that again it gets reinforced when you're putting on a live stream that you put in hours and hours and hours of preparation on and you don't know whether or not the person who actually was the author of the work is going to be able to show up or not because these are busy people and then they do that's an active inference moment that's a kind of a dedication to a philosophy and I think that's where Carl said there's a lot of stuff that still has to be worked out by the philosophers well I continue along with Thomas and Dalton to try to work out what the formalisms and the operations are so I think an eye on each is kind of where we are today there's probably less room for people to say no it's one or the other and I think there's a lot more room to say well if we put this and this and this together I wonder what kind of smoothie that derives hmm that's my that's my sense of where we are at the start of 2023. it was like in the beginning of the year if only we could just pin it down is active inference realist or is active inference this way right if we could just get it on the table on procrustes bed then I would be able to approach it I'm too busy right now I'm too overwhelmed right now it's too complex right now it's too ambiguous right now it's changing too fast right now if you could just stabilize it in this one dimension right then I'll meet you and I'll be ready to talk and out of that environment the doorway was dialogue and getting different perspectives together beginner intermediate and Beyond it is becoming a growing ecosystem of Education research applications and service where it's not just that there can be the coexistence of different perspectives it's that there are there must be and we love that it seems it seems a bit counter-intuitive but the if there is a nature part to one of the things we might Discover at the end of 2023 that are that there's more pockets of feral and Wild behavior and activity as opposed to less that the more times people come and say is there a packaged

ver is there a Disneyland version of this that I can use as my entry point perhaps what they'll discover is there's there's an opportunity you know you when you hit your ball out of bounds in Costa Rica there's signs that says do not find your ball there are scorpions there are snakes oh and there are crocodiles crocodiles don't just inhabit the water areas they literally hang out in the forest and so you respect that right unless you're going to properly protect your your nine iron Ain't Gonna Save You is my point but but there's a there's a there's a piece of this that's that's highly groomed highly formalized and I think there's also a piece of this that from a from a behavioral active standpoint we might find a year from now is a lot more enabling is a lot more feral is a lot more who who who wants to go into those marginal areas and and really poke around a little bit and I think again what people might find is that active inferences into Shield but it is an anchor it provides the related stabilities that people can anchor to so that they can then go out and play with those edges or the non-edges they can function in ways that they probably didn't imagine if they believed that active inference was a ladder to something specific so again there's a creative side of this that I think a year from now again I I know that there's a bunch of people in business that want to do capture work around this and there's a bunch of scientists that want to get confirmation and I get that I get what their goals are and I respect them but I also think there's a creativity piece to this that's just percolating just below the surface and that when it's given a chance to run its course it's gonna it it could be amazing in terms of what people generate it made me think a shield is not a sarcophagus an anchor is not a pillar driven into the ocean floor even something that is stabilizing and defensive must have a loose tethering and an optimal grasp if it's better to be stabilized why not just put the pillar down like that is One Direction of thought it may be valid locally or at a meso scale situationally we're not looking to resolve these questions in the abstract there may be a case where you do want to get inside the iron sarcophagus and hide but when we are out there figuring and balancing that with figuring it out we need to also see those two in conversation or in a dialectic so it's not like we can just figure it such that we're going to de-risk and Achieve out there it's not the work itself it isn't even interestingly enough the exact preparation for the work itself out there but also if we're only out there so much happens that we lose valuable opportunities for pre-flection and reflection and we lose those opportunities to figure things out while the iron is hot and so we grow and we act infer and we serve less than we could which again may not be a bad thing and maybe no lives are lost for it but how to read and speak between the lines and make sure that the active inference Institute is accessible to all language and mathematical backgrounds as our usage of the active inference

ontology and the formalisms becomes increasingly sophisticated as demands from industry from professionalized research and all of these other sectors become increasingly intense as well how will we be able to provide services to our community and Niche neither exactly square with what their priors are because if it did they could have done it themselves nor do we want to come from a different wavelength or planet where the material are failing to make impact because they're not accessible applicable and rigorous here's an interesting thing to think about as as one of the people who's spent countless hours growing this this idea I don't think back in January of this year you could have taken the words from majeed where he was talking about partial encapsulation and envisioned a direct application to the idea that the active inference lab had to be only partially encapsulated in order for it to morph into the uh active inference these are the things that we upon reflection can look back and go well we were talking about it as an idea but here's an example of where it actually became true and we couldn't have I didn't know at the time that the active inference lab was going to turn into the Institute that's what actively inferring allows for that's what's really fascinating and again I'll say this if you really want to learn go through an exercise like you and I just went through today because it gives you the opportunity to say oh by the way if you want some real evidence without pre-staging the idea that I want to prove this I want to prove that there's a partial encapsulation just live it out migrate through it go through that space time come out the other side look back on it and go wow I guess what we were talking about uh 11 months and 26 days ago in fact could turn out to be pretty prescient because there's no way we could become an Institute today if we didn't leave a Gap open for that possibility so it's just really interesting again if you want to if you want active inference to be a repeat of what you have already been habituated to you can make it that or you can stick around and kind of go only mackerel it wasn't just in theory in fact it materialized in the ways that the theory said it could not would and that's the bridge to everywhere hmm we put up our canonical final slide thank you Dean for for this stream and for all of the others and everyone who came on if you're listening this far thank you for your regime of attention we are looking to make 2023 and years Beyond it better and this was a great year so it's no slight on where we've been it's just a spiral of developments any Dimension or Divergence from your highest preference is either feedback you can provide the or engagement that you can yourself make happen you want to see a certain kind of stream happen you want to see another language other than English you want to see a different kind of educational material become accessible either provided as feedback or let's figure out how to make it happen with your engagements we'll go out there

anytime wow no word can close the year so we'll just leave it at that thank you Dean thank
you all right until next time my friend until next time all right ciao